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Firms as Liquidity Providers: Evidence from the 2007–2008 Financial Crisis

Contents

1 Big picture

- **Research question.** When bank credit becomes scarce, do nonfinancial firms with strong liquidity positions provide liquidity to other firms through trade credit?
- **Main idea.** Trade credit can act as a substitute source of financing when banks and financial markets contract. Cash-rich suppliers may use their liquidity to support customers that face financing constraints.
- **Empirical setting.** The paper studies U.S. public firms around the first phase of the 2007–2008 financial crisis, focusing especially on the period from Q3:2007 to Q2:2008.
- **Core outcome.** The main supply-side outcome is accounts receivable divided by sales, interpreted as trade credit provided by suppliers.
- **Demand-side outcome.** The main demand-side outcome is accounts payable divided by cost of goods sold, interpreted as trade credit used by customers.
- **Main finding.** Firms with high precrisis liquidity increased trade credit provision during the financial phase of the crisis, while less liquid firms did not.
- **Who receives liquidity?** Constrained clients, clients with limited liquidity, and clients in industries dependent on external finance increase their use of trade credit, especially when matched with liquid suppliers.
- **Mechanism.** The evidence supports a redistribution view of trade credit: firms with better access to funds redistribute liquidity to firms with worse access to funds through supplier–customer relationships.
- **Dynamic result.** Trade credit provision helped during the early financial phase of the crisis, but the role of trade credit was limited once the crisis became more severe after Lehman and demand collapsed.
- **Economic takeaway.** Corporate cash has a precautionary value not only for the firm holding it, but also for its trading partners. Liquid firms can partly insure their customers when institutional credit is scarce.

2 Incremental contribution of the paper

- **From bank-credit shocks to inter-firm credit.** The paper studies how firms respond to a negative bank-credit supply shock by reallocating liquidity through trade credit.
- **Supplier–customer matched evidence.** The paper goes beyond firm-level accounts receivable by using a matched sample of suppliers and major customers. This allows the authors to control simultaneously for supply and demand factors.
- **Precrisis liquidity as treatment intensity.** Liquidity is measured before the crisis, typically at Q2:2006, reducing concerns that liquidity choices during the crisis are themselves responses to client distress.

- **Demand-control design.** Some specifications use client-quarter fixed effects, comparing different suppliers to the same client in the same quarter. This is a strong way to absorb time-varying client demand for trade credit.
- **Multiple liquidity measures.** The paper distinguishes cash, excess cash, line-of-credit availability, and combined liquidity measures, showing that both cash and precommitted credit lines matter.
- **Supply and demand sides of trade credit.** The analysis studies both accounts receivable and accounts payable. This two-sided evidence helps connect liquidity provision by suppliers to liquidity use by clients.
- **Aggregate shock rather than idiosyncratic distress.** Much prior trade-credit work focuses on idiosyncratic shocks or cross-sectional financing constraints. This paper studies an economy-wide credit contraction.
- **Precautionary cash motive.** The paper provides a concrete reason firms may hold cash: cash lets them preserve valuable customer relationships during financial stress.

3 Data and measurement

3.1 Baseline Compustat sample

The baseline data come from Compustat quarterly files for U.S.-incorporated nonfinancial firms.

- **Main period.** The main precrisis and crisis sample runs from Q3:2005 to Q2:2008.
- **Extended period.** The long-run dynamic analysis extends the sample to Q2:2010.
- **Firm restrictions.** The sample excludes financial firms and observations with invalid assets, cash, sales, accounts receivable, extreme asset growth, or extreme sales growth.
- **Size filters.** The sample excludes very small firms, including firms with market capitalization below $50 million and book assets below $10 million.
- **Baseline size.** The whole Compustat sample contains 24,733 firm-quarter observations and 2,250 firms in the main descriptive table.

Interpretation. The baseline sample is broad and representative of public nonfinancial firms, but it does not directly identify the clients receiving trade credit. It motivates the more precise matched-sample analysis.

3.2 Supplier–customer matched sample

The matched sample uses Compustat's Customer Segment File, which reports major customers that account for more than 10 percent of a supplier's sales.

- **Unit of observation.** The key unit is a supplier–customer relation in a quarter.
- **Matching.** Customer names are matched to Compustat firms using string matching and manual matching.

- **Final regression sample.** The main matched sample contains 9,368 supplier–customer pair-quarter observations with non-missing variables.
- **Advantage.** The matched data allow the authors to control for client characteristics and, in some specifications, client-quarter fixed effects.
- **Limitation.** The data only identify major public customers. They do not reveal the exact accounts receivable owed by each customer to each supplier.

Interpretation. The matched sample is not fully representative of every commercial relationship, but it is powerful because it directly links suppliers to important customers.

3.3 Liquidity variables

The paper measures suppliers' liquidity before the crisis to reduce simultaneity concerns.

- **Cash/assets.** Cash reserves scaled by total assets, measured at Q2:2006.
- **Excess cash/assets.** Actual cash minus predicted cash from an optimal cash-holdings model, scaled by total assets.
- **Line-of-credit dummy.** An indicator for whether the firm has a line of credit.
- **LOC limit/assets.** The total credit-line limit scaled by assets.
- **LOC undrawn balance/assets.** The unused portion of the line of credit scaled by assets.
- **Liquidity 1.** Cash plus total LOC limit, scaled by assets.
- **Liquidity 2.** Cash plus undrawn LOC balance, scaled by assets.

Interpretation. Cash and lines of credit are imperfect substitutes. Cash is immediately available, while credit lines provide committed external liquidity subject to covenants and bank willingness to honor commitments.

3.4 Trade credit variables

- **Trade credit provided.** Accounts receivable divided by sales, denoted informally as $AR/Sales$.
- **Trade credit used.** Accounts payable divided by cost of goods sold, denoted informally as $AP/COGS$.
- **Scaling by flows.** The paper scales by sales or cost of goods sold to account for changes in real activity during crises.
- **Relation-level interpretation.** In the matched sample, the dependent variable is still the supplier's total accounts receivable or the client's total accounts payable, not the exact bilateral trade-credit amount.

Interpretation. The trade-credit measures are imperfect but economically meaningful. A rise in accounts receivable relative to sales means the supplier is financing a larger share of sales through customer credit.

3.5 Financial constraints and external finance dependence

The paper uses both firm-level and industry-level measures of constraints.

- **Kaplan–Zingales index.** A firm-level index intended to capture financing constraints.
- **Whited–Wu index.** Another firm-level financial-constraint index.
- **Payout ratio.** An inverse proxy for constraints: firms that pay out more are typically less constrained.
- **No-rating dummy.** Firms without a long-term debt rating are treated as more constrained.
- **Line-of-credit measures.** LOC limit and unused LOC balances are inverse measures of constraints.
- **External finance dependence, EFD.** An industry-level Rajan–Zingales-style measure of the fraction of investment not financed by internal cash flow.

Interpretation. Firm-level constraint measures may be endogenous. Industry-level EFD helps because it is predetermined and less tied to the firm's own crisis behavior.

4 Methodology and empirical specifications

4.1 Difference-in-differences logic

The empirical strategy compares trade credit before and after the onset of the crisis as a function of firms' precrisis liquidity.

- The shock is the financial phase of the 2007–2008 crisis.
- The treatment intensity is the supplier's liquidity measured one year before the crisis.
- The key question is whether liquid firms change trade credit provision differently from illiquid firms during the crisis.
- The preferred interpretation is that the crisis raises the value of alternative financing, and liquid suppliers are better able to provide it.

4.2 Baseline accounts-receivable specification

The main supplier-side specification can be written as:

$$AR_{it} = \beta_{ij} + \beta_1 Crisis_t + \beta_2 (Crisis_t \times LIQ_{i,t^*}) + \beta_3 X_{i,t-1} + \beta_4 X_{j,t-1} + \beta_5 w_{ijt} + \alpha_{ij} + \varepsilon_{ijt}.$$

- AR_{it} is the supplier's accounts receivable divided by sales.
- $Crisis_t$ equals one from Q3:2007 to Q2:2008.
- LIQ_{i,t^*} is supplier liquidity measured before the crisis, usually Q2:2006.

- X_i and X_j are supplier and client controls.
- w_{ijt} is the importance of the client to the supplier, measured as sales to that client divided by supplier sales.
- α_{ij} is a supplier–customer pair fixed effect.

Main coefficient. β_2 measures whether liquid suppliers increased trade credit provision more during the crisis.

4.3 Client-quarter fixed effects

A stronger matched-sample specification replaces pair fixed effects with client-quarter fixed effects.

- This compares different suppliers selling to the same client in the same quarter.
- It absorbs all time-varying client demand for trade credit.
- The coefficient on supplier liquidity is identified from variation across suppliers facing the same customer-quarter.

Interpretation. If liquid suppliers still provide more trade credit after absorbing client-quarter demand, the result is more likely to reflect supply-side liquidity provision.

4.4 Accounts-payable specification

The paper also studies trade credit from the client's perspective:

$$AP_{jt} = \beta_{ij} + \beta_1 Crisis_t + \beta_2 (LIQ_{i,t^*} \times Crisis_t) + \beta_3 Z_{i,t-1} + \beta_4 Z_{j,t-1} + \beta_5 w_{jit} + \alpha_{ij} + \varepsilon_{ijt}.$$

- AP_{jt} is the client's accounts payable divided by cost of goods sold.
- The key liquidity variable remains the supplier's precrisis liquidity.
- A positive coefficient means clients matched with more liquid suppliers increase trade-credit debt more during the crisis.

Interpretation. This provides the demand-side mirror image of the accounts-receivable evidence.

4.5 Constraint tests

To test whether constrained firms demand more trade credit, the paper adds client constraint measures interacted with the crisis dummy.

- A positive crisis interaction for constraint measures means constrained clients increase accounts payable during the crisis.
- Supplier liquidity interactions remain in many specifications to show that constrained clients receive trade credit especially when their suppliers can provide it.

- Supplier-quarter fixed effects are used in some specifications to absorb unobserved time-varying supplier characteristics.

Interpretation. Trade credit is most valuable when clients are constrained and suppliers have liquidity.

4.6 Long-run dynamics and performance

The paper extends the analysis through Q2:2010 by separating the crisis into three periods.

- **Crisis 2007.** Q3:2007–Q2:2008, the early financial phase.
- **Crisis 2008.** Q3:2008–Q2:2009, the post-Lehman severe crisis phase.
- **Post-crisis.** Q3:2009–Q2:2010.

The paper also examines performance by relating changes in trade credit provision to return on assets, return on sales, EBITDA, and net profit margin.

Interpretation. Liquidity provision is helpful but not unlimited. Once the crisis deepens and cash is depleted, firms shift toward rebuilding liquidity.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether precrisis liquidity predicts differential trade-credit behavior after a large negative credit-supply shock. The evidence is strongest for the claim that liquid suppliers increase trade credit relative to illiquid suppliers during the early financial crisis.

5.2 Why the 2007–2008 phase matters

The authors focus on Q3:2007–Q2:2008 because this period is more plausibly dominated by a credit-supply contraction. After Lehman, the crisis also includes a sharp collapse in demand, which makes it harder to separate supply and demand effects.

5.3 Supply versus demand

A central concern is that accounts receivable may rise because clients cannot or will not pay, not because suppliers choose to provide liquidity. The paper addresses this in several ways.

- Liquidity is measured before the crisis.
- The supplier–customer matched sample controls for client characteristics.
- Client-quarter fixed effects compare suppliers to the same client in the same quarter.
- A placebo test around the 9/11 demand shock does not produce the same pattern.

- Accounts payable increases for clients matched with liquid suppliers, consistent with actual liquidity transfers.

5.4 Why external finance dependence is useful

External finance dependence is measured at the industry level rather than at the firm level. Low-EFD suppliers are less dependent on external funds and therefore more able to provide liquidity. High-EFD clients are more likely to need liquidity. The strongest pattern is liquidity flowing from low-EFD suppliers to high-EFD clients.

5.5 Why lines of credit matter

Cash alone is not the full liquidity position. Firms with access to unused lines of credit can draw on committed bank financing and pass liquidity through to customers. The results show that combined cash-plus-LOC measures often have strong predictive power.

5.6 Limitations

- The matched sample observes only major public customers disclosed in Compustat.
- The exact bilateral trade-credit amount between a supplier and a customer is not observed.
- The empirical design is observational, not randomized.
- The crisis affects both credit supply and real demand, especially after Q3:2008.
- Measures such as accounts receivable and accounts payable can reflect both intentional credit extension and delayed payment.

Interpretation. The paper does not prove every dollar of accounts receivable is voluntary liquidity insurance, but the pattern across supplier liquidity, client constraints, matched fixed effects, placebo tests, and performance results strongly supports the liquidity-provision interpretation.

6 Tables and figures

6.1 Table 1: Descriptive statistics

Read:

- Table 1 reports summary statistics for the whole Compustat sample and the supplier–customer matched sample.
- The whole sample contains 24,733 firm-quarter observations; the matched sample contains 9,368 pair-quarter observations.
- The median supplier in the matched sample looks broadly similar to the median Compustat firm.

- Clients in the matched sample are much larger than typical Compustat firms, which is expected because suppliers disclose major customers.
- The table reports cash, excess cash, line-of-credit availability, and the combined liquidity measures used later in the regressions.

Economic message: The matched supplier–customer sample is selective but still informative. It gives the paper a setting where supplier liquidity and client characteristics can be analyzed together.

6.2 Table 2: Baseline supplier liquidity and trade credit provision

Read:

- Table 2 estimates supplier-level regressions in the whole Compustat sample.
- The dependent variable is accounts receivable over sales.
- The interaction of the crisis dummy with cash is positive and significant, showing that cash-rich firms provide more trade credit during the crisis.
- Excess cash, cash plus LOC limit, and cash plus unused LOC balances also predict greater trade credit provision.
- The effect is stronger for rated firms and for firms in low external-finance-dependence industries.
- The placebo test around the 9/11 demand shock does not replicate the main finding.

Economic message: The baseline evidence suggests a supply effect: firms with more precrisis liquidity are better able to finance customers when bank credit tightens.

Table 1
Descriptive statistics.

This table contains summary statistics of key variables in the two samples used in the paper: Compustat firms and supplier–customer relations. All statistics are calculated from July 1, 2005 to June 30, 2008, except for the liquidity variables, which are measured exactly once per firm or relation, at the second quarter of 2006. The last two columns contain the percentage of Compustat firms that are covered in the supplier–customer pairs sample, and the ratio of the medians [means] of each variable in the Compustat sample to the respective values in the paired sample. The distributions for the customer–supplier sample used in the last two columns are calculated at the firm-quarter level (i.e., each firm-quarter appears only once per supplier or client). * Denotes (mean A/mean B) instead of medians.

	Sample A. Whole compustat (firm-quarters)				Sample B. Customer–supplier pairs (pair-quarters)				Comparison between sample A and B	
Variable name	Number of observations	Mean	Median	Standard deviation	Number of observations	Mean	Median	Standard deviation	Compustat coverage (percentage of observations)	Median A/ median B
Supplier variables										
Account receivables/ sales	24,733	0.612	0.590	0.419	9,368	0.642	0.607	0.341	23.74%	0.983
Accounts payable/cost of goods sold	24,648	0.649	0.447	0.819	9,334	0.666	0.490	0.813	23.75%	0.905
Assets (Millions of US dollars)	24,733	3,824	566	14,237	9,368	2,974	512	10,007	23.74%	1.037
Age (years)	24,733	20.70	16.00	13.13	9,368	19.37	15.00	12.58	23.74%	1.024
Property plant and equipment/assets	24,733	0.245	0.172	0.218	9,368	0.194	0.136	0.189	23.74%	1.206
Net profit margin	24,733	0.245	0.368	1.184	9,368	0.258	0.385	1.044	23.74%	0.956
Sales growth	24,733	0.001	0.027	0.250	9,368	−0.005	0.027	0.279	23.74%	0.977
Net worth/assets	24,733	0.377	0.392	0.279	9,368	0.414	0.435	0.283	23.74%	0.915
Total debt/assets	24,733	0.197	0.158	0.206	9,368	0.182	0.131	0.201	23.74%	1.235
Tobin's *q*	24,733	2.034	1.605	1.331	9,368	2.101	1.638	1.386	23.74%	0.987
Liquidity variables (measured at Q2:2006)										
Cash/assets	2,250	0.199	0.116	0.214	998	0.253	0.183	0.236	22.40%	0.631
Excess cash/assets	1,999	0.129	0.052	0.229	909	0.180	0.118	0.255	23.01%	0.437
Line of credit (LOC) dummy	1,694	0.530	1.000	0.499	744	0.461	0.000	0.499	22.02%	1.183*
LOC limit/assets	1,694	0.054	0.000	0.099	744	0.042	0.000	0.086	22.02%	1.272*
LOC undrawn balance/assets	1,694	0.019	0.000	0.054	744	0.018	0.000	0.052	22.02%	1.067*
(LOC limit + cash)/assets	1,694	0.249	0.193	0.208	744	0.293	0.241	0.221	22.02%	0.866*
(LOC undrawn balance + cash)/assets	1,694	0.214	0.149	0.214	744	0.269	0.210	0.232	22.02%	0.816*
Client variables										
Assets (millions of US dollars)	24,733	3,824	566	14,237	9,368	62,360	28,019	115,868	13.62%	0.674
Age (years)	24,733	20.70	16.00	13.13	7,266	31.51	36.00	13.38	8.68%	0.841
Sales growth	24,733	0.001	0.027	0.250	9,368	0.010	0.023	0.202	13.62%	1.015
Total debt/assets	24,733	0.197	0.158	0.206	9,368	0.211	0.197	0.148	13.62%	0.747
No rating (dummy)	24,733	0.665	1.000	0.472	9,368	0.114	0.000	0.318	13.62%	2.652*
Market share	24,733	0.062	0.007	0.146	9,360	0.227	0.163	0.228	13.60%	0.129
Net profit margin	24,660	0.248	0.368	1.185	9,368	0.286	0.288	1.143	13.66%	1.164
Cash/assets	24,733	0.192	0.111	0.208	9,368	0.113	0.068	0.128	13.62%	1.562
Kaplan-Zingales index	21,034	0.865	0.799	0.793	6,074	0.848	0.796	0.509	10.50%	0.880
Whited-Wu index	23,530	−0.304	−0.296	0.097	8,711	−0.472	−0.501	0.079	13.53%	0.669
Payout ratio	24,598	0.002	0.000	0.005	9,040	0.004	0.003	0.005	13.02%	0.645*
Weight (sales to client/ total supplier sales)					9,368	0.161	0.130	0.149		

Table 1: Descriptive statistics for Compustat firms and supplier–customer pairs.

Table 2
Supplier's liquidity and trade credit provision during the crisis.

The table presents estimates from panel regressions explaining firm-level quarterly trade credit provided for quarters with an end date from July 1, 2005 to June 30, 2008. The dependent variable is accounts receivable over sales. Crisis is an indicator that equals one from the third quarter of 2007 to the second quarter of 2008. The top row indicates the liquidity measure that is interacted with the crisis dummy in each regression: Cash reserves in Columns 1 and 5–9, Excess cash in Column 2, and total liquidity ratios in Columns 3, 4, and 10. Cash reserves is the ratio of cash to total assets. Excess cash is the residual cash to total assets and is defined relative to the model of optimal cash holdings as presented in Dittmar and Mahrt-Smith (2007), estimated over the period 1995–2004. Liquidity 1 is the ratio of cash reserves plus total amount in lines of credit, scaled by assets. Liquidity 2 is the ratio of cash reserves plus unused amount in lines of credit, scaled by assets. Cash reserves, Excess cash and lines of credit (LOCs) are measured at the end of the last fiscal quarter ending before July 1, 2006. In Columns 5 and 6 the sample is split into mutually exclusive subsamples according to firms' availability of a bond rating, and in Columns 7 and 8 the sample is split into mutually exclusive subsamples according to the firms' external finance dependence (EFD). EFD is the industry-median proportion of investment not financed by cash flow from operations. The last row of the table provides the F-statistic and the p-value associated to the test of equality of the interaction coefficients of the mutually exclusive subsets. The table also presents placebo tests using the demand crisis following September 11, 2001 (Columns 9 and 10). In this case the estimation sample consists of quarterly data from the last quarter of year 2000 and the third quarter of 2002, Cash and LOC variables are measured at the third quarter of year 2000, and Crisis is an indicator that equals to one from the fourth quarter of 2001 to the third quarter of 2002. All specifications control for firms' characteristics, which include size, age, tangibility, net profit margin, sales growth, net worth, total debt, Tobin's *q*, and dummies for bond ratings. All specifications include firm fixed effects. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

	Cash, excess cash, lines of credit, rating, and external finance dependence								Placebo test	
	Cash reserves	Excess cash	Liquidity 1 (Cash + LOC limit)	Liquidity 2 (Cash + Unused LOC)	Cash reserves, rated	Cash reserves, unrated	Cash reserves, low EFD	Cash reserves, high EFD	Cash reserves	Liquidity 2 (Cash + Unused LOC)
Independent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Crisis	−0.015***	−0.008***	−0.024***	−0.019***	−0.023***	−0.013**	−0.015***	−0.016***	−0.016***	−0.023***
	[0.004]	[0.004]	[0.005]	[0.004]	[0.004]	[0.006]	[0.006]	[0.005]	[0.004]	[0.004]
Crisis × Liquidity	0.024*	0.021*	0.046***	0.028**	0.130***	0.017	0.061***	−0.004	−0.043***	−0.365
	[0.013]	[0.012]	[0.014]	[0.014]	[0.029]	[0.016]	[0.018]	[0.018]	[0.014]	[0.518]
Log of total assets	0.062***	0.042***	0.066***	0.067***	0.068***	0.061***	0.033***	0.079***	0.079***	0.076***
	[0.007]	[0.007]	[0.008]	[0.008]	[0.009]	[0.009]	[0.010]	[0.010]	[0.007]	[0.009]
Log of age	−0.043***	−0.041***	−0.047***	−0.046***	−0.045***	−0.050***	0.020	−0.056***	−0.041***	−0.044***
	[0.013]	[0.013]	[0.015]	[0.015]	[0.013]	[0.017]	[0.018]	[0.017]	[0.011]	[0.013]
Property, plant, and equipment over assets	−0.185***	−0.221***	−0.269***	−0.266***	−0.011	−0.264	0.185	−0.319	0.100	0.113
	[0.038]	[0.039]	[0.042]	[0.042]	[0.045]	[0.052]	[0.065]	[0.048]	[0.041]	[0.048]
Net profit margin	−0.038***	−0.037***	−0.028***	−0.027***	−0.088***	−0.037***	−0.069***	−0.034***	−0.071***	−0.057***
	[0.002]	[0.002]	[0.003]	[0.003]	[0.017]	[0.003]	[0.005]	[0.003]	[0.004]	[0.005]
Sales growth	−0.174***	−0.168***	−0.177***	−0.177***	−0.162***	−0.178***	−0.173***	−0.178***	−0.234***	−0.201***
	[0.005]	[0.006]	[0.006]	[0.006]	[0.008]	[0.007]	[0.008]	[0.007]	[0.006]	[0.007]
Net worth over assets	−0.128***	−0.131***	−0.118***	−0.119***	−0.237***	−0.116***	0.192***	−0.280***	0.003	0.026
	[0.02]	[0.02]	[0.022]	[0.023]	[0.033]	[0.024]	[0.029]	[0.026]	[0.020]	[0.026]
Debt over assets	−0.266***	−0.219***	−0.298***	−0.298***	−0.235***	−0.282***	−0.054***	−0.345***	−0.021	−0.043
	[0.026]	[0.027]	[0.030]	[0.030]	[0.037]	[0.034]	[0.037]	[0.036]	[0.028]	[0.034]
Tobin's *q*	−0.008***	−0.008***	−0.008***	−0.009***	0.004	−0.010	−0.001	−0.011	0.004	0.003
	[0.002]	[0.002]	[0.002]	[0.002]	[0.005]	[0.003]	[0.003]	[0.003]	[0.001]	[0.002]
Rating dummies	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.063	0.060	0.064	0.064	0.078	0.062	0.082	0.069	0.086	0.074
Number of observations	24,733	22,234	19,432	19,432	8,233	16,500	9,563	15,170	26,710	15,007
Number of firms	2,250	1,999	1,694	1,694	739	1,511	876	1,374	3,002	1,589
F statistic					5.957		5.727			
p-Value					0.015		0.017			

Table 2: Supplier liquidity and trade credit provision during the crisis.

6.3 Tables 3 and 4: Matched supplier–customer evidence

Read Table 3:

- Table 3 uses the supplier–customer matched sample.
- The interaction of crisis and supplier cash remains positive after controlling for supplier and client characteristics.
- Suppliers provide more trade credit to high-EFD clients than to low-EFD clients.
- Low-EFD suppliers are the main providers of additional liquidity.
- The client-time fixed effects specification strengthens the evidence by comparing suppliers to the same client in the same quarter.
- The 9/11 placebo again does not produce a significant effect.

Read Table 4:

- Table 4 examines alternative supplier liquidity measures in the matched sample.
- Supplier cash, excess cash, cash plus LOC limit, and cash plus unused LOC balances are all positively associated with trade credit provision during the crisis.
- Client liquidity tends to reduce the need for supplier credit.
- Client-quarter fixed effects confirm that more liquid suppliers provide more trade credit even when client demand is absorbed flexibly.

Economic message: The matched-sample results are the heart of the paper. Liquidity provision is not just a firm-level correlation; it appears within actual commercial relationships and survives strong controls for client demand.

Table 3
Trade credit provision and supplier's liquidity controlling for client s characteristics.

This table presents estimates from panel regressions explaining firm-level quarterly trade credit provided for quarters with an end date from July 1, 2005 to June 30, 2008 using a sample of firms that report their main customers. Each observation represents a supplier–client relation. The dependent variable is the supplier's accounts receivable over sales. Suppliers' cash reserves are measured prior to the onset of the crisis and interacted with the crisis dummy in all regressions. In Columns 4 and 5 we divide the sample into two mutually exclusive groups according to client's external finance dependence (low or high), and in Columns 6 and 7 the sample is divided into mutually exclusive groups according to supplier's EFD (low or high). In Column 9 the crisis refers to the 2001 demand crisis following the events of September 11, 2001 (as defined in Table 2). All specifications except for the first control for suppliers' characteristics (defined in Table 2). Columns 2–7 and 9 also control for client's characteristics, which include size, sales growth, total debt, and a dummy for not having a long-term debt rating. Specifications 1–7 and 9 include pairs fixed effects, and specification 8 contains client-quarter fixed effects and supplier cash to assets ratio. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. Standard errors are clustered at the pair level.

Independent variable	No controls (1)	Client controls (2)	Client's EFD (3)	Client low EFD (4)	Client high EFD (5)	Supplier low EFD (6)	Supplier high EFD (7)	Client-time fixed effects (8)	Placebo (9)
Crisis	−0.006	−0.01	−0.014*	−0.003	−0.025**	−0.014	−0.011		−0.034**
	[0.007]	[0.008]	[0.008]	[0.011]	[0.011]	[0.011]	[0.010]		[0.015]
Supplier variables									
Crisis × Supplier's cash	0.059**	0.066**	0.058**	0.017	0.103**	0.056*	0.057	0.133***	0.062
	[0.030]	[0.030]	[0.029]	[0.028]	[0.043]	[0.034]	[0.040]	[0.043]	[0.050]
Log of total assets		−0.018	−0.017	−0.028	−0.012	−0.032	−0.009	−0.020***	0.033
		[0.021]	[0.021]	[0.027]	[0.028]	[0.027]	[0.028]	[0.004]	[0.050]
Log of age		0.052***	0.048***	0.014	0.244***	0.242***	0.022	0	−0.023
		[0.016]	[0.016]	[0.016]	[0.076]	[0.077]	[0.016]	[0.007]	[0.036]
Property, plant, and equipment over assets		−0.179	−0.189	−0.193	−0.136	0.03	−0.309	−0.409***	0.565***
		[0.132]	[0.134]	[0.168]	[0.173]	[0.158]	[0.188]	[0.032]	[0.215]
Net profit margin		0.005	0.006	−0.023*	0.016	0.003	0.001	−0.020***	−0.016
		[0.014]	[0.014]	[0.013]	[0.015]	[0.007]	[0.014]	[0.005]	[0.020]
Sales growth		−0.096***	−0.091***	−0.148***	−0.073***	−0.044**	−0.138***	−0.106***	−0.140***
		[0.015]	[0.015]	[0.029]	[0.017]	[0.021]	[0.020]	[0.014]	[0.020]
Net worth over assets		0.074	0.067	−0.099	0.230***	0.265***	−0.044	−0.286***	0.200**
		[0.056]	[0.056]	[0.066]	[0.086]	[0.079]	[0.071]	[0.023]	[0.081]
Debt over assets		0.059	0.042	−0.190**	0.237**	0.139	−0.01	−0.402***	0.232*
		[0.076]	[0.077]	[0.089]	[0.116]	[0.103]	[0.098]	[0.032]	[0.119]
Tobin's q		0.009	0.009	0.003	0.018	0.007	0.011	−0.022***	−0.005
		[0.006]	[0.006]	[0.004]	[0.012]	[0.007]	[0.008]	[0.004]	[0.004]
Client variables									
Importance		−0.197	−0.036	−0.037	−0.363*	−0.031	−0.263	−0.140***	0.003
		[0.121]	[0.028]	[0.061]	[0.205]	[0.07]	[0.161]	[0.029]	[0.029]
Log of total assets		−0.029	0.005	−0.068	−0.031	−0.101	0.009		0.013
		[0.027]	[0.009]	[0.060]	[0.026]	[0.062]	[0.025]		[0.194]
Sales growth		0.006	0.008	0.008	0.005	0.011	−0.002		−0.027**
		[0.009]	[0.046]	[0.018]	[0.011]	[0.016]	[0.012]		[0.011]
Debt over assets		−0.005	−0.014	0.031	−0.013	0.007	−0.031		−0.03
		[0.047]	[0.017]	[0.078]	[0.060]	[0.062]	[0.067]		[0.092]
No rating dummy		−0.012	0.119	−0.008	−0.025	0.026	−0.019		−0.004
		[0.018]	[0.098]	[0.026]	[0.027]	[0.034]	[0.021]		[0.039]
Market share		0.121	−0.009**	0.453***	−0.132	0.033	0.148		−0.115
		[0.099]	[0.004]	[0.146]	[0.096]	[0.131]	[0.114]		[0.115]
Net profit margin		−0.009**	−0.208*	0.031	−0.009***	−0.012***	−0.008		0.012**
		[0.004]	[0.121]	[0.019]	[0.004]	[0.003]	[0.006]		[0.005]
Cash over assets		−0.108	−0.104	−0.027	−0.118	−0.002	−0.128		0.112
		[0.095]	[0.096]	[0.091]	[0.152]	[0.116]	[0.125]		[0.138]
Client's EFD × Crisis 2007			0.051***						
			[0.020]						
Supplier rating dummies	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pair fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Client-quarter fixed effects	No	No	No	No	No	No	No	Yes	No
Number of observations	9,360	9,360	9,252	4,527	4,725	3,161	6,199	9,360	5,920
Number of pairs (client-quarters)	1,341	1,341	1,326	654	672	434	907	−3,363	1,280
R-squared	0.002	0.038	0.038	0.068	0.051	0.046	0.055	0.135	0.049
F statistic				2.86		0.003			
p-Value				0.091		0.958			

Table 3: Supplier cash and trade credit provision controlling for client characteristics.

Table 4
Suppliers' liquidity, clients' liquidity, and trade credit provision during the crisis.

This table presents estimates from panel regressions explaining firm-level quarterly trade credit provided for quarters with an end date from July 1, 2005 to June 30, 2008 using a sample of firms that report their main customers. Each observation represents supplier–client relations. The dependent variable is the supplier's accounts receivable over sales. Suppliers' liquidity variables are cash in Column 1, Excess cash in Column 2, Cash + line of credit (LOC) limit in Column 3, and Cash + Unused LOC balances in Columns 4 and 5. Liquidity is measured prior to the onset of the crisis and interacted with the crisis dummy in all regressions. Specifications in Panel A contain suppliers' characteristics (as defined in Table 2) and client's characteristics, which include size, sales growth, total debt, a dummy for not having a long-term debt rating, client external finance dependence (EFD), and client liquidity. Client liquidity is measured as Cash in Columns 1–4 and as Cash + Unused LOC balances in Column 4. Specifications in Panel B contain client-quarter fixed effects and supplier controls. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. Standard errors are clustered at the pair level.

Panel A: Supplier–customer pairs fixed effects

Independent variable	Cash reserves (1)	Excess cash (2)	Liquidity 1 (Cash + LOC limit) (3)	Liquidity 2 (Cash + Unused LOC) (4)	Liquidity 2 (Cash + Unused LOC) (5)
Crisis	−0.014*	−0.013**	−0.022**	−0.017*	−0.018
	[0.008]	[0.006]	[0.010]	[0.009]	[0.012]
Crisis × Supplier liquidity	0.058**	0.063**	0.089**	0.074**	0.074*
	[0.029]	[0.029]	[0.037]	[0.035]	[0.045]
Client liquidity	−0.104	−0.146*	−0.100	−0.099	−0.002***
	[0.096]	[0.079]	[0.119]	[0.119]	[0.001]
Measure of client liquidity	Cash	Cash	Cash	Cash	Liquidity 2
Supplier controls	Yes	Yes	Yes	Yes	Yes
Client controls	Yes	Yes	Yes	Yes	Yes
Pair fixed effects	Yes	Yes	Yes	Yes	Yes
R-squared	0.038	0.042	0.052	0.051	0.057
Number of observations	9,252	8,444	7,069	7,069	5,169
Number of pairs	1,326	1,199	987	987	677

Panel B: Client-quarter fixed effects

Independent variable	Cash reserves (6)	Excess cash (7)	Liquidity 1 (Cash + LOC limit) (8)	Liquidity 2 (Cash + Unused LOC) (9)
Supplier liquidity	−0.212***	−0.263***	−0.079**	−0.123***
	[0.030]	[0.030]	[0.035]	[0.036]
Crisis × Supplier liquidity	0.133***	0.159***	0.108*	0.132**
	[0.043]	[0.054]	[0.063]	[0.061]
Supplier controls	Yes	Yes	Yes	Yes
Client controls	No	No	No	No
Client-quarter fixed effects	Yes	Yes	Yes	Yes
R-squared	0.135	0.135	0.172	0.173
Number of observations	9,360	8,695	7,315	7,315
Number of client-quarters	3,363	3,342	2,985	2,985

Table 4: Supplier liquidity, client liquidity, and trade credit provision.

6.4 Tables 5 and 6: Debt constraints and accounts payable

Read Table 5:

- Table 5 replaces liquidity measures with leverage measures.
- Suppliers with more short-term debt or net short-term debt reduce trade credit provision during the crisis.
- Long-term debt and total debt are not statistically significant in the same way.
- The result is consistent with refinancing risk limiting a supplier's ability to provide customer credit.

Read Table 6:

- Table 6 switches to the client side and uses accounts payable over cost of goods sold as the dependent variable.
- Clients matched with more liquid suppliers increase trade-credit debt during the crisis.
- The interaction is positive for supplier cash, supplier excess cash, supplier LOC access, and combined liquidity measures.
- The evidence connects supplier liquidity provision to client use of trade credit.

Economic message: Liquidity flows through the trade-credit channel. Suppliers with refinancing pressure pull back, while clients with liquid suppliers borrow more through accounts payable.

Table 5
Supplier's leverage and trade credit provision during the crisis.

This table presents estimates from panel regressions explaining firm-level quarterly trade credit provided for quarters with an end date from July 1, 2005 to June 30, 2008 using a sample of firms that report their main customers. Each observation represents a supplier–client relationship. The dependent variable is the supplier's accounts receivable over sales. Suppliers' leverage variables are measured prior to the onset of the crisis and interacted with the crisis dummy in all regressions (as liquidity variables in Table 2). Leverage variables are Short-term debt to assets in Column 1, Long-term debt to assets in Column 2, Total debt to assets in Column 3, and Net short-term debt (short-term debt net of cash) to assets in Column 4. All specifications contain suppliers characteristics (as defined in Table 2), client characteristics (as in Table 4), and pair fixed effects. ***, **, and * indicate that the coefficient is significant at the 1%, 5%, and 10% level, respectively. Standard errors are clustered at the pair level.

Independent variable	Short-term debt (1)	Long-term debt (2)	Total debt (3)	Net short-term debt (4)
Crisis	0.013**	0.008	0.013	−0.006
	[0.006]	[0.008]	[0.008]	[0.007]
Crisis × Leverage measure	−0.151*	−0.008	−0.027	−0.063**
	[0.089]	[0.039]	[0.035]	[0.027]
Firm (supplier) controls	Yes	Yes	Yes	Yes
Client controls	Yes	Yes	Yes	Yes
Pair fixed effects	Yes	Yes	Yes	Yes
Number of observations	9,292	9,325	9,274	9,292
R-squared	0.038	0.036	0.037	0.038
Number of pairs	1,320	1,332	1,316	1,320

Table 5: Supplier leverage and trade credit provision during the crisis.

Table 6
Client's debt in trade credit and supplier's liquidity.

This table presents estimates from panel regressions explaining firm-level quarterly debt in trade credit for quarters with an end date from July 1, 2005 to June 30, 2008. Each observation represents a supplier–client relation. The dependent variable is the client's accounts payable over cost of goods sold. All specifications include client and supplier controls. Client controls include size (measured in assets), age, current assets to assets ratio, sales growth, total debt over assets, and cash over assets. Suppliers' characteristics include size, sales growth, total debt over assets, and net profit margin. All but the first specification also include the interaction of supplier's liquidity (measured one year previous to the start of the crisis) with the crisis dummy, as in Tables 2–6. Liquidity is measured as Cash reserves in Column 2, Excess cash in column 3, LOC (line of credit) dummy in Column 4, Cash plus LOC limit in Column 5, and Cash plus undrawn LOC balances in Column 6. Supplier's weight is the ratio of purchases from the given supplier to total cost of goods sold. All specification include relation fixed effects and a constant. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. Standard errors are clustered at the pair level.

Independent variable	None (1)	Supplier's cash (2)	Supplier's excess cash (3)	Supplier's LOC dummy (4)	Supplier's liquidity 1 (Cash+LOC Limit) (5)	Supplier's liquidity 2 (Cash+Unused LOC) (6)
Crisis	−0.027***	−0.034***	−0.020***	−0.034***	−0.038***	−0.037***
	[0.005]	[0.005]	[0.006]	[0.005]	[0.007]	[0.007]
Crisis × Supplier's liquidity		0.140**	0.071**	0.048**	0.096***	0.096**
		[0.060]	[0.032]	[0.019]	[0.036]	[0.038]
Client's log of total assets	−0.131***	−0.135***	−0.138***	−0.134***	−0.133***	−0.133***
	[0.028]	[0.028]	[0.029]	[0.029]	[0.029]	[0.029]
Client's log of age	0.038***	0.031***	0.035***	0.044***	0.036***	0.036***
	[0.011]	[0.011]	[0.011]	[0.012]	[0.011]	[0.011]
Client's current assets	−0.491***	−0.444***	−0.444***	−0.428***	−0.423***	−0.424***
	[0.074]	[0.067]	[0.069]	[0.069]	[0.069]	[0.069]
Client's sales growth	−0.210***	−0.185***	−0.187***	−0.193***	−0.192***	−0.192***
	[0.029]	[0.020]	[0.021]	[0.021]	[0.021]	[0.021]
Client's debt over assets	−0.150	−0.179**	−0.190**	−0.221**	−0.221**	−0.219**
	[0.092]	[0.088]	[0.093]	[0.094]	[0.094]	[0.094]
Client's cash over assets	−0.579***	−0.550***	−0.573***	−0.583***	−0.553***	−0.553***
	[0.065]	[0.065]	[0.068]	[0.069]	[0.069]	[0.070]
Supplier's log of assets	0.025**	0.016	0.019*	0.018*	0.018*	0.018*
	[0.012]	[0.011]	[0.011]	[0.011]	[0.011]	[0.011]
Supplier's sales growth	0.000	0.001	0.000	0.000	0.000	0.000
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
Supplier's debt over assets	0.064**	0.070**	0.068**	0.082**	0.081**	0.081**
	[0.030]	[0.032]	[0.034]	[0.034]	[0.034]	[0.034]
Supplier's net profit margin	0.001	−0.001	−0.001	−0.001	−0.002	−0.002
	[0.003]	[0.002]	[0.002]	[0.002]	[0.002]	[0.002]
Supplier's weight	0.985***	0.904***	0.922***	0.837**	0.842**	0.843**
	[0.341]	[0.328]	[0.333]	[0.367]	[0.369]	[0.369]
Pair fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.063	0.056	0.057	0.053	0.053	0.053
Number of observations	14,815	14,331	13,853	13,670	13,670	13,670
Number of pairs	2,407	2,222	2,121	2,097	2,097	2,097

Table 6: Client accounts payable and supplier liquidity.

6.5 Tables 7 and 8: Client constraints and crisis dynamics

Read Table 7:

- Table 7 examines whether constrained clients increase trade-credit debt.
- Most constraint measures have the expected sign during the crisis.
- Whited–Wu, low payout, no rating, low LOC availability, unused LOC, and external finance dependence all show evidence of greater trade-credit use by constrained firms.
- Supplier cash remains important in many specifications.
- Supplier-quarter fixed effects absorb unobservable time-varying supplier characteristics.

Read Table 8:

- Table 8 extends the analysis from Q3:2005 to Q2:2010.
- In the whole sample, the positive liquidity-provision effect is concentrated in Crisis 2007 and weakens or reverses later.

- In the matched sample, liquidity provision remains positive during the 2008 crisis once client controls are included.
- By the post-crisis period, the liquidity interaction largely disappears.

Economic message: Trade credit can partially absorb a temporary financial shock, but it is not an unlimited substitute for bank credit during a severe and persistent crisis.

Table 7
Credit constraints and debt in trade credit.

This table presents estimates from panel regressions explaining firm-level quarterly debt in trade credit for quarters with an end date from July 1, 2005 to June 30, 2008. Each observation represents a supplier–client relation. The dependent variable is the client's accounts payable over cost of goods sold. All specifications include client controls which include size (measured in assets), age, current assets to assets ratio, sales growth, total debt over assets, cash over assets, and the importance of the supplier to the given client, measured as purchases from that supplier over total cost of goods sold. Specifications in Panel A also include suppliers' controls, which include size, sales growth, total debt over assets, and net profit margin, and the interaction of supplier's cash reserves (measured one year previous to the start of the crisis) with the crisis dummy, as in Tables 2–6. Specifications in Panel A include relation fixed effects and a constant. Specifications in Panel B control for supplier-quarter fixed effects. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. Standard errors are clustered at the pair level.

Panel A: Debt in trade credit, client constraints, and supplier's cash

Independent variable	Kaplan-Zingales (1)	Whited-Wu (2)	Payout ratio (3)	No rating dummy (4)	LOC limit (5)	Unused LOC (6)	External finance dependence (7)
Crisis	−0.039***	0.216***	−0.018***	−0.036***	−0.033***	−0.024***	−0.063***
	[0.013]	[0.033]	[0.007]	[0.005]	[0.006]	[0.005]	[0.008]
Crisis × Client's constraints	0.006	0.517***	−0.171***	0.040***	−0.248**	−0.928***	0.282***
	[0.013]	[0.070]	[0.057]	[0.015]	[0.118]	[0.197]	[0.035]
Crisis × Supplier's cash	0.140*	0.024	0.121**	0.115*	0.164**	0.164**	0.091
	[0.075]	[0.064]	[0.059]	[0.065]	[0.070]	[0.070]	[0.059]
Firm (client) controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Supplier controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pair fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.064	0.061	0.057	0.056	0.059	0.064	0.064
Observations	8,998	14,047	14,212	14,331	12,527	12,527	14,331
Number of pairs	1,412	2,163	2,202	2,222	1,954	1,954	2,222

Panel B: Debt in trade credit and credit constraints, controlling for time varying unobservable supplier characteristics

	Kaplan-Zingales (8)	Whited-Wu (9)	Payout ratio (10)	No rating dummy (11)	LOC limit (12)	Unused LOC (13)	External finance dependence (14)
Client's constraints	−0.087**	0.005	0.535***	−0.096***	0.904***	2.390***	0.579***
	[0.040]	[0.279]	[0.098]	[0.034]	[0.169]	[0.251]	[0.082]
Crisis × Constraint	0.039	0.719***	−0.453***	0.087	−0.315	−1.198***	0.271**
	[0.055]	[0.268]	[0.172]	[0.068]	[0.248]	[0.334]	[0.127]
Firm (client) controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Supplier-quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.273	0.178	0.176	0.156	0.215	0.231	0.187
Number of observations	10,510	16,192	16,394	16,553	14,514	14,514	14,815
Number of supplier-quarters	7,727	10,396	10,451	10,523	9,720	9,720	9,704

Table 7: Credit constraints and debt in trade credit.

Table 8
Supplier's cash and liquidity provision over the crisis episode.

This table presents estimates from panel regressions explaining firm-level quarterly trade credit provided for quarters with an end date from July 1, 2005 to June 30, 2010. The dependent variable is accounts receivable over sales. Whole sample specifications are estimated using Compustat firms and include supplier fixed effects. Supplier–customer sample specifications are estimated using relation data and include pair fixed effects. The top row indicates the liquidity measure that is interacted with the crisis dummies in each regression: Cash reserves in columns 1 and 4, Excess cash in Columns 2 and 5, and Cash+Unused balance in line of credit (LOC) in Columns 3 and 6. Cash and LOC variables are measured at the second quarter of year 2006. Crisis 2007 is an indicator that equals to one from the third quarter of 2007 to the second quarter of 2008. Crisis 2008 is an indicator that equals to one from the third quarter of 2008 to the second quarter of 2009. Post-crisis is an indicator that equals to one from the third quarter of 2009 to the second quarter of 2010. All other variables are defined in Tables 2 and 4. All specifications control for suppliers' characteristics (as in Table 2). Supplier–customer sample specifications also control for client characteristics (as in Table 4). All specifications include firm fixed effects. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

	Whole sample			Supplier–customer sample		
Independent variable	Cash reserves (1)	Excess cash (2)	Liquidity 2 (3)	Cash reserves (4)	Excess cash (5)	Liquidity 2 (6)
Crisis 2007	−0.015***	−0.008**	−0.019***	−0.012	−0.007	−0.018*
	[0.004]	[0.004]	[0.005]	[0.008]	[0.007]	[0.010]
Crisis 2008	−0.031***	−0.027***	−0.034***	−0.023***	−0.017**	−0.023**
	[0.004]	[0.004]	[0.005]	[0.009]	[0.008]	[0.010]
Post-crisis	−0.014***	−0.016***	−0.016***	−0.004	−0.004	−0.007
	[0.005]	[0.004]	[0.005]	[0.014]	[0.013]	[0.016]
Crisis 2007 × Liquidity	0.023*	0.024*	0.031**	0.092	0.099***	0.108***
	[0.014]	[0.013]	[0.015]	[0.035]	[0.035]	[0.041]
Crisis 2008 × Liquidity	−0.005	−0.013	0.006	0.108***	0.105***	0.110**
	[0.015]	[0.014]	[0.016]	[0.040]	[0.038]	[0.045]
Post-crisis × Liquidity	−0.059***	−0.033**	−0.046***	−0.002	−0.020	−0.007
	[0.016]	[0.015]	[0.016]	[0.054]	[0.055]	[0.055]
Supplier controls	Yes	Yes	Yes	Yes	Yes	Yes
Client controls	No	No	No	Yes	Yes	Yes
Supplier fixed effects	Yes	Yes	Yes	No	No	No
Pair fixed effects	No	No	No	Yes	Yes	Yes
R-squared	0.060	0.060	0.058	0.033	0.035	0.042
Number of observations	38,236	34,325	31,375	11,566	10,614	9,188
Number of firms (pairs)	2,249	1,998	1,693	(1,046)	(964)	(769)

Table 8: Supplier liquidity provision over the crisis episode.

6.6 Table 9 and Appendix Table A1: Performance and robustness

Read Table 9:

- Table 9 studies whether liquidity provision is associated with firm performance.
- Supplier performance improves when suppliers increase trade credit provision during the crisis, using precrisis cash as an instrument for the change in receivables.
- Improvements are especially visible in return on sales, EBITDA, and net profit margin.
- Clients that benefit from suppliers' increased accounts receivable also show better performance in several measures, although net profit margin effects are weaker.

Read Appendix Table A1:

- Appendix Table A1 reports robustness checks for sample windows, cash measurement dates, and scaling choices.
- The positive interaction between Crisis 2007 and cash generally survives alternative denominators and sample periods.
- The robustness checks are reported for both the whole sample and the supplier–customer sample.

Economic message: The performance evidence suggests that liquidity provision is not merely involuntary delayed payment. Liquid suppliers appear to benefit from preserving customer relationships, and the main finding is robust across multiple specification choices.

Table 9
Liquidity provision and firm performance.

This table presents estimates from panel regressions explaining firm-level quarterly performance measures. Dependent variables are return on assets (Columns 1–3), return on sales (Columns 4–6), earnings before interest, taxes, depreciation and amortization (EBITDA, Columns 7–9), and net profit margin (Columns 10–12). For each performance measure, we run the estimations over three different time frames: the 2007–2008 crisis, the complete 2007–2009 crisis episode, and from the start of the credit crunch (Q3:2007) to one year after the crisis (Q2:2010). Crisis is an indicator that equals to one from the third quarter of 2007 to the second quarter of 2008 (Columns 1, 4, 7, and 10). Crisis is an indicator that equals to one from the third quarter of 2007 to the second quarter of 2009 (Columns 2, 5, 8, and 11). Crisis is an indicator that equals to one from the third quarter of 2007 to the second quarter of 2010 (Columns 3, 6, 9, and 12). ΔAR/S is the change of the average accounts receivable to sales ratio in Q3:2006–Q2:2007 relative to the average ratio during Q3:2007–Q2:2008. In Panel A, we instrument ΔAR/S with supplier's precrisis cash stocks. All specifications control for firms' characteristics, which include size, age, tangibility, sales growth, net worth, Tobin's q, total debt, and ratings dummies. All specifications include firm fixed effects and a constant. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

	Return on assets			Return on sales			EBITDA			Net profit margin		
Independent variable	Q2:2008 (1)	Q2:2009 (2)	Q2:2010 (3)	Q2:2008 (4)	Q2:2009 (5)	Q2:2010 (6)	Q2:2008 (7)	Q2:2009 (8)	Q2:2010 (9)	Q2:2008 (10)	Q2:2009 (11)	Q2:2010 (12)
Panel A: Supplier performance												
Crisis	−0.004***	−0.009***	−0.008***	−0.046**	−0.051***	−0.065***	−0.004***	−0.007***	−0.009***	−0.061***	−0.054***	−0.073***
	[0.001]	[0.001]	[0.001]	[0.021]	[0.017]	[0.018]	[0.001]	[0.001]	[0.002]	[0.017]	[0.015]	[0.016]
Predicted [Crisis × (ΔAR/S)]	0.017	0.028*	0.036**	1.473***	0.732**	1.103***	0.093***	0.145***	0.204***	1.831***	1.841***	2.233***
	[0.013]	[0.015]	[0.015]	[0.377]	[0.339]	[0.355]	[0.032]	[0.039]	[0.049]	[0.302]	[0.291]	[0.326]
Supplier controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	25,437	32,655	36,149	25,437	32,655	36,149	24,404	31,446	37,458	25,358	32,562	38,702
Number of firms	2,047	2,047	2,047	2,047	2,047	2,047	2,033	2,037	2,040	2,047	2,047	2,047
Cragg-Donald Wald *F*	73.13	82.77	83.04	73.13	82.77	83.04	14.89	19.06	21.30	71.76	81.87	81.95
Panel B: Client performance												
Crisis	−0.002	−0.005***	−0.005***	−0.010	−0.021**	−0.016	−0.001	−0.001*	−0.002***	−0.005**	−0.004	−0.003
	[0.001]	[0.001]	[0.001]	[0.007]	[0.010]	[0.012]	[0.001]	[0.001]	[0.001]	[0.002]	[0.009]	[0.007]
Crisis × ΔSupplier AR/S	0.012***	0.010***	0.010***	0.067***	0.054**	0.056*	0.001	0.004**	0.004***	−0.001	0.010	0.006
	[0.003]	[0.003]	[0.003]	[0.019]	[0.025]	[0.029]	[0.002]	[0.002]	[0.002]	[0.005]	[0.022]	[0.018]
Client controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Supplier controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	2,139	2,898	3,283	2,139	2,898	3,283	2,091	2,838	3,607	2,138	2,897	3,674
Number of firms	197	210	210	197	210	210	196	210	223	197	210	223

Table 9: Liquidity provision and firm performance.

Table A1
Robustness checks on sample, cash measurement, and dependent variable denominator.

This table presents several specifications for validation purposes. The dependent variable is accounts receivable over sales. The sample consists of quarterly data. We use the whole sample of suppliers in Panel A and the supplier–customer sample in Panel B. Crisis 2007 is an indicator that equals to one from the third quarter of 2007 to the second quarter of 2008. Crisis 2007 × Cash refers to the interaction of the crisis indicator with firm's cash. All specifications control for firms' characteristics, which include size, age, tangibility, net profit margin, sales growth, net worth, Tobin's *q*, total debt, and rating dummies. In addition, in Panel B we include client characteristics as in Table 4. Tangibility, net profit margin, net worth, and total debt are scaled by noncash assets in Column 1, by assets net of accounts receivables in Column 2, and by total assets in Columns 3–11.The starting points of the sample are quarters starting on July 1, 2005 for Columns 1, 2, and 6–8; July 1, 2004 for Columns 3–5 and July 1, 2006 for Columns 9–11. In all columns, the ending points of the sample period is the quarter ending on June 30, 2008. Cash is measured at the end of the last fiscal quarter ending before July 1 of year 2006 in Columns 1, 2, 3, 6, and 9; year 2005 in Columns 4, 7, and 10; and year 2004 in Columns 5, 8, and 11. All specifications in Panel A include firm fixed effects, and in Panel B they include pair fixed effects. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

	Sample: Q3:2005 to Q2:2008		Sample: Q3:2004 to Q2:2008			Sample: Q3:2005 to Q2:2008			Sample: Q3:2006 to Q2:2008		
	Non-cash assets	Non-receivables assets	Sales			Sales			Sales		
Independent variable	Q2:2006 (1)	Q2:2006 (2)	Q2:2006 (3)	Q2:2005 (4)	Q2:2004 (5)	Q2:2006 (6)	Q2:2005 (7)	Q2:2004 (8)	Q2:2006 (9)	Q2:2005 (10)	Q2:2004 (11)
Panel A: Whole sample estimations											
Crisis 2007	−0.016***	−0.015***	−0.016***	−0.016***	−0.015***	−0.015***	−0.011***	−0.012***	−0.016***	−0.014***	−0.016***
	[0.003]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]
Crisis 2007 × Cash	0.012***	0.022*	0.030**	0.027**	0.018	0.024*	0.005	0.005	0.028**	0.010	0.022*
	[0.003]	[0.012]	[0.013]	[0.013]	[0.013]	[0.013]	[0.013]	[0.013]	[0.014]	[0.014]	[0.013]
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.068	0.056	0.062	0.060	0.062	0.063	0.059	0.063	0.061	0.056	0.061
Number of observations	24,739	24,707	33,264	34,346	34,462	24,733	24,991	24,801	16,093	15,972	15,875
Number of Firms	2,250	2,245	2,256	2,455	2,608	2,250	2,443	2,423	2,242	2,231	2,219
Panel B: Supplier–customer estimations											
Crisis 2007	−0.006	−0.006	−0.010	−0.014*	−0.010	−0.010	−0.009	−0.005	−0.011	−0.012*	−0.008
	[0.008]	[0.008]	[0.008]	[0.007]	[0.007]	[0.008]	[0.008]	[0.007]	[0.007]	[0.007]	[0.007]
Crisis 2007 × Cash	0.018*	0.044*	0.049	0.057**	0.037	0.066**	0.058**	0.035	0.051*	0.055**	0.031
	[0.011]	[0.025]	[0.030]	[0.026]	[0.024]	[0.030]	[0.028]	[0.025]	[0.028]	[0.026]	[0.024]
Pairs fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.039	0.037	0.033	0.033	0.033	0.038	0.037	0.038	0.047	0.044	0.043
Number of observations	9,360	9,356	12,458	12,944	12,981	9,360	9,483	9,424	6,236	6,184	6,155
Number of pairs	1,341	1,339	1,545	1,656	1,706	1,341	1,430	1,418	1,186	1,179	1,172

Appendix Table A1: Robustness checks on sample, cash measurement, and denominator.

7 Short summary

- The paper studies whether firms provide liquidity to other firms through trade credit during the 2007–2008 financial crisis.
- It uses U.S. public-firm data and a matched supplier–customer sample based on major customers disclosed in Compustat.
- The main supplier-side outcome is accounts receivable over sales.
- The main client-side outcome is accounts payable over cost of goods sold.
- Firms with high precrisis cash increase trade credit provision during Q3:2007–Q2:2008.
- Excess cash and credit-line availability strengthen the effect.
- The result is strongest for suppliers that are less dependent on external finance and for clients that are more dependent on external finance.
- Client-quarter fixed effects show that more liquid suppliers provide more credit to the same client in the same quarter.
- Constrained clients increase trade-credit debt during the crisis.
- Suppliers with short-term debt reduce trade credit provision, consistent with refinancing risk limiting liquidity supply.
- The 9/11 placebo test does not generate the same pattern, supporting the interpretation that the 2007 shock was primarily a credit-supply shock.
- Trade credit provision is valuable but limited: the effect weakens after the crisis deepens and firms begin rebuilding cash.
- The paper identifies a precautionary motive for corporate cash: cash helps firms support important trading partners when external finance dries up.

8 Conclusion

8.1 Core conclusion

Garcia-Appendini and Montoriol-Garriga show that nonfinancial firms can act as liquidity providers during a credit crunch. Firms with more precrisis liquidity increase trade credit extended to customers during the early 2007–2008 financial crisis, and constrained clients increase trade credit use when matched with liquid suppliers.

8.2 Economic intuition

When bank credit is plentiful, customers can rely more heavily on banks and financial markets. When bank credit tightens, trade credit becomes a substitute source of liquidity. Suppliers with cash or unused credit lines face a lower opportunity cost of lending to customers and can preserve valuable commercial relationships by financing them temporarily.

8.3 Incremental contribution revisited

The paper's key contribution is to connect corporate liquidity management, trade credit, and the financial crisis in a matched supplier–customer design. It shows not only that liquid firms behave differently, but also that liquidity appears to flow from liquid suppliers to constrained clients through actual trading relationships.

8.4 Implications for corporate cash policy

The findings strengthen the precautionary-savings view of cash. Cash reserves are not only useful for financing the firm's own investment or survival. They also help a firm protect its customer base, maintain sales relationships, and potentially improve performance during financial stress.

8.5 Implications for financial-crisis transmission

The paper shows that bank-credit shocks can be partially absorbed by nonbank credit channels. Trade credit acts as a decentralized liquidity backstop. However, because suppliers' liquidity is finite, this channel cannot fully offset a severe and persistent financial crisis.

8.6 Limitations

The empirical design is observational and relies on precrisis liquidity as a source of differential exposure. Accounts receivable and accounts payable are aggregate balance-sheet items, not exact bilateral credit flows. The matched sample covers only disclosed major customers. Demand effects become especially difficult to separate after Lehman, when the crisis affects both credit supply and real activity.

8.7 Takeaway

The enduring takeaway is that firms can be liquidity intermediaries. During the early financial crisis, cash-rich suppliers helped finance constrained customers through trade credit. This provides a real channel through which corporate liquidity buffers can stabilize supply chains when banks and markets pull back.